

Hydro MPC

Installation and operating instructions



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English (GB) Installation and operating instructions

Original installation and operating instructions

These installation and operating instructions apply to the Grundfos Hydro MPC booster systems.

Sections 1-4 give the information necessary to be able to install the product in a safe way.

Sections 5-16 give important information about the product as well as information on service and fault finding.

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Read this document before you install the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document

1.1 Warnings against hazards involving risk of death or personal injury



DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.



WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.



CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The text accompanying the three hazard symbols DANGER, WARNING and CAUTION is structured in the following way:



SIGNAL WORD

Description of hazard

Consequence of ignoring the warning.
- Action to avoid the hazard.

1.2 Other important notes



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.



Tips and advice that make the work easier.

2. Receiving the product

2.1 Transporting the product

Depending on the size, the booster system is delivered in an open wooden box or wooden or cardboard box designed for transport by forklift truck or a similar vehicle.

The forks of the forklift truck must be at least 2 m long.



The Hydro MPC booster systems with CR 120 or CR 150 pumps are secured by means of transport straps. Do not remove these transport straps until the booster system has been installed.

3. Installing the product

Before installing the product, check the following:

- The booster system corresponds to the order.
- All visible parts are intact.

3.1 Location

Install the booster system in a well-ventilated room to ensure sufficient cooling of the control cabinet and pumps.



Hydro MPC is only designed for indoor installation. Do not expose the product to direct sunlight.

Place the booster system with a 1 m clearance in front and on the two sides for inspection and removal.

3.2 Mechanical installation

3.2.1 Pipes

Arrows on the pump base show the direction of flow of water through the pump.

The pipes connected to the booster system must be of adequate size.

Connect the pipes to the manifolds of the booster system. Either end can be used. Apply sealing compound to the unused end of the manifold, and fit the screw cap. For manifolds with flanges, fit a blanking flange with gasket.

To achieve optimum operation and minimise noise and vibration, it may be necessary to consider vibration dampening of the booster system.

Noise and vibration are generated by the rotations in the motor and pump and by the flow in pipes and fittings. The effect on the environment is subjective and depends on correct installation and the state of the other parts of the system.

If booster systems are installed in blocks of flats or the first consumer on the line is close to the booster system, we recommend that you fit expansion joints on the inlet and outlet pipes to prevent that vibrations are transmitted through the pipes.

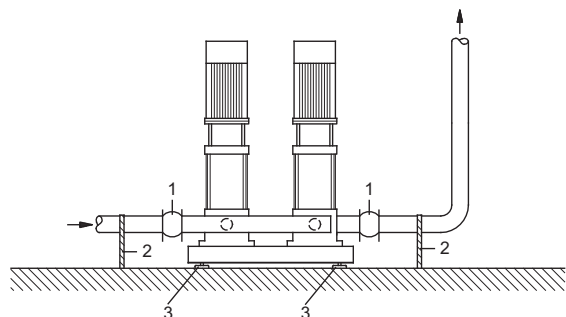


Fig. 1 Example showing the position of expansion joints, pipe supports and machine shoes

Pos.	Description
1	Expansion joint
2	Pipe support
3	Machine shoe



Expansion joints, pipe supports and machine shoes shown in fig. 1 are not included in a standard booster system.

Tighten all nuts before startup.

Fasten the pipes to parts of the building to ensure that they cannot move or be twisted.

3.2.2 Foundation

We recommend that you install the booster system on a plane and rigid concrete foundation which is heavy enough to provide permanent support for the entire system. The foundation must be capable of absorbing any vibration, normal strain or shock.



The weight of a concrete foundation must be 1.5 times the weight of the booster system.

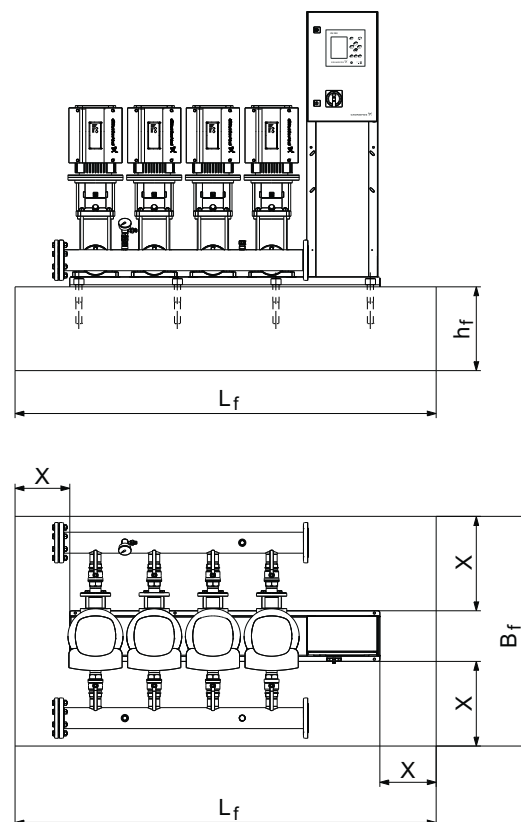


Fig. 2 Foundation

The minimum height of the foundation, h_f , is calculated as follows:

$$h_f = \frac{m_{\text{pump}} \times 1.5}{L_f \times B_f \times \delta_{\text{concrete}}}$$

The density δ of concrete is usually taken as 2200 kg/m³.

3.2.3 Vibration dampers

To prevent the transmission of vibrations to buildings, we recommend that you isolate the booster system foundation from building parts by means of vibration dampers.

The right damper varies from installation to installation, and a wrong damper may increase the vibration level. Vibration dampers must therefore be sized by the supplier. If the booster system is installed on a base frame with vibration dampers, always fit expansion joints on the manifolds. This is important to prevent the booster system from "hanging" in the pipes.

3.2.4 Expansion joints

Fit expansion joints for these reasons:

- to absorb expansions or contractions in the pipes caused by changing liquid temperature
- to reduce mechanical strains in connection with pressure surges in the pipes
- to isolate mechanical structure-borne noise in the pipes (only rubber bellows expansion joints).



Do not install expansion joints to compensate for inaccuracies in the pipes such as centre displacement of flanges.

Fit expansion joints at a distance of minimum 1 to 1 1/2 times the nominal flange diameter from the manifold on the inlet as well as on the outlet side. This prevents the development of turbulence in the expansion joints, resulting in better inlet conditions and a minimum pressure loss on the pressure side.

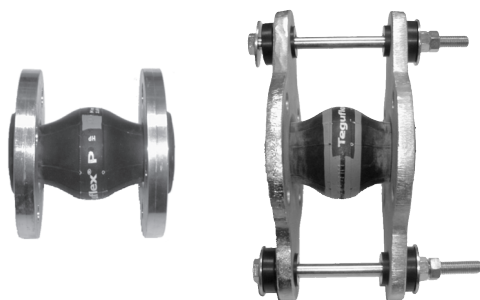


Fig. 3 Examples of rubber bellows expansion joints without and with limiting rods

Expansion joints with limiting rods can be used to minimise the forces caused by the expansion joints. We always recommend that you use expansion joints with limiting rods for flanges larger than DN 100.

Anchor the pipes so that they do not stress the expansion joints and the pump. Follow the supplier's instructions and pass them on to advisers or pipe installers.

3.2.5 Prefilling of diaphragm tank, if applicable

If a diaphragm tank is connected to the system, prefill the tank with nitrogen to this pressure:

Hydro MPC-E and -F: 0.7 x the setpoint.

Hydro MPC-S: 0.9 x the setpoint.



Use nitrogen to avoid corrosion.

3.3 Electrical installation



The electrical installation must be carried out by an authorised person in accordance with local regulations and the relevant wiring diagram.

- The electrical installation of the booster system must comply with enclosure class IP54.
- Check that the power supply and frequency correspond to the values stated on the nameplate.
- Make sure that the conductor cross-section meets the specifications in the wiring diagram.
- If the system cannot be installed with the supply disconnecting device located minimum 0.6 m above service level (ground level), install the system with an external "supply disconnecting device" made according to EN 60204-1, paragraph 5.3.2. The system must be provided with a means permitting it to be locked in the OFF (isolated) position. Based on a risk assessment performed by the installer or end-user, the device must be installed in a position according to EN 60204-1, paragraph 5.3.4. The system must be connected to an external emergency stop device or an emergency switch-off device according to the requirements of EN ISO 13850.

4. Starting up the product

After having carried out the mechanical and electrical installation described in sections 3.2 *Mechanical installation* and 3.3 *Electrical installation*, proceed as follows:

1. Switch on the power supply.
2. Wait for the first display to appear.
3. The first time CU 352 is switched on, a startup wizard guides the user through the basic settings.
4. Follow the instructions in each display.
5. When the wizard is completed, check that all pumps are set to "Auto" in the menu "Status".
6. Go to the menu "Operation".
7. Select the operating mode "Normal" and press [OK].
8. The system is now ready for operation.

4.1 Handling the product

The Hydro MPC booster systems with CR 120 or 150 pumps have eyebolts in the base frame. See fig. 4.

During handling, the lifting point must always be above the centre of gravity of the booster system.

Each lifting strap must be at least 3 m long.

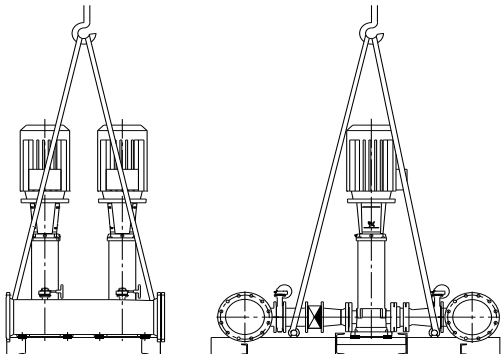


Fig. 4 Correct lifting of Hydro MPC XL

CAUTION

Overhead load



- Minor or moderate personal injury
- When lifting the booster system, do not use the eyebolts of the motors.
 - Do not lift the booster system by the manifolds.
 - Do not stand on the manifolds.

CAUTION

Crushing of feet



- Minor or moderate personal injury
- When lifting the booster system, do not use the eyebolts of the motors.
 - Do not lift the booster system by the manifolds.
 - Do not stand on the manifolds.

When lifting the booster system, only use suitable lifting equipment that is in good condition and approved for the weight. The weight is stated on the nameplate of the booster system.



Do not use chains for lifting booster systems with CR 120 or CR 150 pumps, as this may damage the motors.

5. Product introduction

5.1 Product description

As standard, the booster systems consist of two to six CR(I)E) or CR(E) pumps connected in parallel and mounted on a common base frame with a control cabinet and all necessary fittings.



A diaphragm tank must be included in some installations.

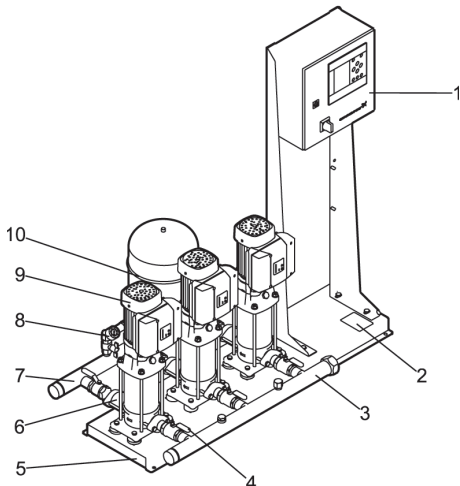


Fig. 5 Hydro MPC booster system

Pos.	Description	Quantity
1	Control cabinet	1
2	Nameplate	1
3	Inlet manifold, stainless steel	1
4	Isolating valve	2 per pump
5	Base frame, stainless steel	1
6	Non-return valve	1 per pump
7	Outlet manifold, stainless steel	1
8	Pressure transmitter and pressure gauge	1
9	Pump	2-6
10	Diaphragm tank	1

5.2 Control variant

The Hydro MPC booster systems are divided into three groups based on the control variant:

Control variant	Description
-E	Two to six electronically speed-controlled pumps. From 0.37 to 22 kW, Hydro MPC-E is equipped with CR(I)E pumps with integrated frequency converter. As from 30 kW, Hydro MPC-E is equipped with CR pumps connected to Grundfos CUE frequency converters (one per pump).
-F	Two to six CR(I) pumps connected to a Grundfos CUE frequency converter. The speed-controlled operation alternates between the pumps.
-S	Two to six mains-operated CR(I) pumps

Design code E-I only uses CR pumps connected to Grundfos CUE frequency converters (one per pump).

See also section 6. *Overview of control variants.*

Hydro MPC always include application-optimised software for setting the booster system to the application in question.

5.3 Identification

5.3.1 Nameplate

The nameplate of the booster system is fitted on the base frame. See position 2 in fig. 6.

The nameplate contains the following information:

- Type: ①
- Model: ②
- Serial No.: ③
- Main supply: ④
- pMax: bar ⑤
- Q Nom / Max: / m³/h ⑦
- Liq. temp.: °C ⑥
- H Nom / Max: / m ⑧
- ⑨
- IP Class: ⑩
- Weight: kg ⑪
- MADE IN GERMANY ⑭
- QR code ⑫
- CE mark ⑬
- GRUNDFOS logo ⑮
- DK - 8850 - Bjerringbro - Denmark ⑮

Fig. 6 Nameplate

Pos.	Description
1	Product type
2	Code for model
3	Serial number
4	Mains supply
5	Maximum operating pressure
6	Liquid temperature
7	Nominal flow rate [m ³ /h]
8	Maximum flow rate [m ³ /h]
9	Technical documents
10	Enclosure class
11	Weight
12	Approval marks
13	QR code
14	Country of origin
15	Company address

5.3.2 Software label

The software label is placed on the back of the CU 352 control unit.

The software label contains the following information:

- 1. Control MPC ①
- 2. C-MPC options ②
- 3. Hydro MPC ③
- 4. H-MPC options ④
- 5. Pump data ⑤
- GRUNDFOS logo
- CONFIGURATION STEPS - PLEASE FOLLOW THE NUMBERS
- 96586126

Fig. 7 Software label

Pos.	Description
1	Control MPC - GSC file number
2	Control MPC options - GSC file numbers
3	Hydro MPC - GSC file number*
4	Hydro MPC options - GSC file numbers*
5	Pump data - GSC file numbers**

* Applies only to booster systems.

** Applies only to CR, CRI, CRN, CRE and CRIE pumps.



A GSC (Grundfos Standard Configuration) file is a configuration data file.

5.4 Type key

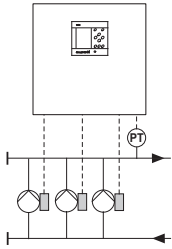
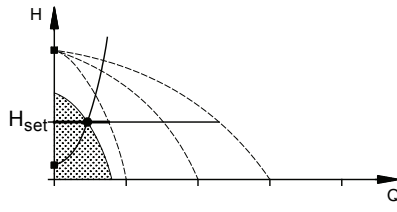
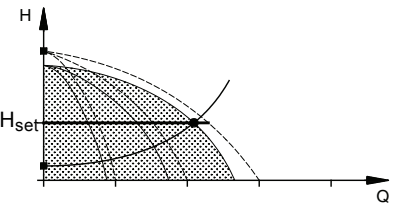
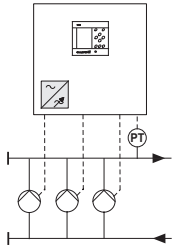
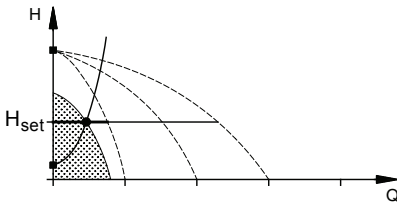
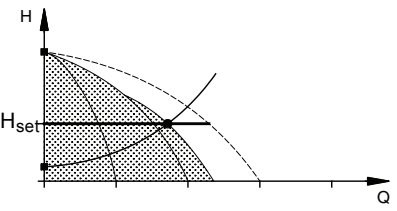
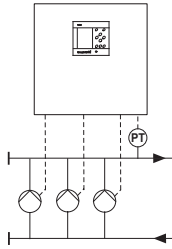
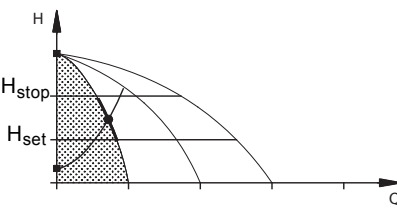
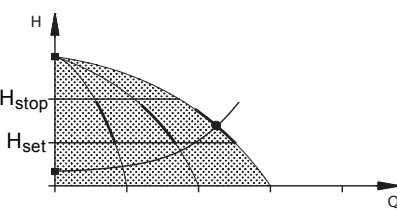
Code	Example	Hydro MPC	-E	6	CRE 150-4-5	U1	A-	A-	A-	ABCD
Type range										
System type										
E	All pumps, E-motor or CUE									
F	Fixed-speed pumps, one CUE									
S	Fixed-speed pumps									
X	Customised-system pumps									
Number of main pumps										
Pump type										
Voltage code										
U1	3 x 380-415 V, N, PE, 50/60 Hz									
U2	3 x 380-415 V, PE, 50/60 Hz									
U3	3 x 380-415 V, N, PE, 50 Hz									
U4	3 x 380-415 V, PE, 50 Hz									
U5	3 x 380-415 V, N, PE, 60 Hz									
U6	3 x 380-415 V, PE, 60 Hz									
U7	1 x 200-240 V, PE, 50/60 Hz									
U8	1 x 200-240 V, N, PE, 50/60 Hz									
U9	3 x 220-240 V, PE, 60 Hz									
UA	3 x 440-480 V, PE, 60 Hz									
UB	1 x 220-240 V, N, PE, 50/60 Hz									
UC	1 x 220-240 V, N, PE, 50 Hz									
UD	3 x 440-480 V, N, PE, 60 Hz									
UX	CSU variant (special voltage rating)									
Design										
A	Systems with the control cabinet mounted on the same base frame as the pumps									
C	Systems with the control cabinet mounted on its own base for floor mounting*									
D	Systems with the control cabinet mounted on its own base frame*									
E	ASEAN design and systems with the control cabinet mounted on the same base frame as the pumps									
F	ASEAN design and systems with the control cabinet centred on the base frame									
G	ASEAN design and systems with the control cabinet mounted on its own base for floor mounting*									
H	ASEAN design and systems with the control cabinet mounted on its own base frame*									
I	ASEAN design and systems with the control cabinet prepared for wall mounting*									
W	Systems with the control cabinet prepared for wall mounting*									
Starting method										
A	E									
B	DOL									
C	SD									
Material combination										
A	Stainless-steel manifold, base frame and standard valves									
B	Stainless-steel manifold, base frames and standard valves									
C	Galvanised-steel manifold, base frame and standard valves									
D	Stainless-steel manifold, galvanised steel base frame and standard valves									
H	Galvanised-steel manifold, base frame painted black and standard valves									
I	Stainless-steel manifold, base frame painted black and standard valves									
X	Customised material combination									
Options										
A	Standard hydraulics									
B	Pilot pump									
C	Bypass									
D	Non-return valve									
E	Elbow manifold									
F	No inlet manifold									
G	Diaphragm tank									
H	Dry-running protection									
I	Repair switch									
J	Redundant sensor									
K	One free position									
L	Two free positions									
M	Three free positions									
N	PN 10 pressure rating									
O	PN 25 pressure rating									
Q	PN 40 pressure rating									
P	Low precharge pressure									
S	Customised variant									
T	Certificate									
U	Undersized motor									
V	Standard controls with options									
W	Customised controls									
X	More than four options									

Design code: E-I is only available in selected countries.

* The control cabinet can be placed up to 2 m from the pumps.

6. Overview of control variants

The table shows examples of systems.

Systems with speed-controlled pumps	Systems with pumps connected to one CUE frequency converter	Systems with mains-operated pumps
Hydro MPC-E	Hydro MPC-F	Hydro MPC-S
Hydro MPC booster system with three CR(I)E pumps.  One CR(I)E pump in operation.  Three CR(I)E pumps in operation.  - Hydro MPC-E maintains a constant pressure through continuous adjustment of the speed of the pumps. - The system performance is adjusted to the demand through cutting in/out the required number of pumps and through parallel control of the pumps in operation. - Pump changeover is automatic and depends on load, operating hours and fault. - All pumps in operation will run at equal speed.	System with three CR pumps connected to one Grundfos CUE frequency converter in the control cabinet. The speed-controlled operation alternates between the pumps.  One CR pump connected to one Grundfos CUE frequency converter in operation.  One CR pump connected to one Grundfos CUE frequency converter and two mains-operated CR pumps in operation.  - Hydro MPC-F maintains a constant pressure through continuous adjustment of the speed of the CR pump connected to the Grundfos CUE frequency converter. The speed-controlled operation alternates between the pumps. - One CR pump connected to the Grundfos CUE frequency converter always starts first. If the pressure cannot be maintained by the pump, one or two mains-operated CR pumps will be cut in. - Pump changeover is automatic and depends on load, operating hours and fault.	System with three mains-operated CR(I) pumps.  One mains-operated CR(I) pump in operation.  Three mains-operated CR(I) pumps in operation.  - Hydro MPC-S maintains an almost constant pressure through cutting in/out the required number of pumps. - The operating range of the pumps will lie between H_{set} and H_{stop} (cut-out pressure). - Pump changeover is automatic and depends on load, operating hours and fault.

7. Operating panel

The operating panel in the front cover of the control cabinet features a display, a number of buttons and two indicator lights. The operating panel enables manual setting and monitoring of the performance of the system.

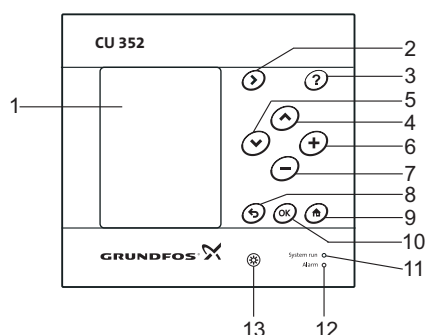


Fig. 8 Operating panel

Pos.	Description
1	Display
2	Arrow to the right
3	Help
4	Up
5	Down
6	Plus
7	Minus
8	Back
9	Home
10	OK
11	Indicator light, operation (green)
12	Indicator light, fault (red)
13	Brightness

7.1 Display

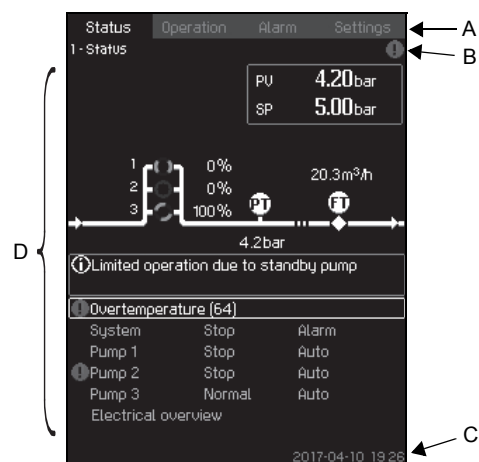


Fig. 9 Display design

7.1.1 Menu line

The menu line (A) is illustrated in fig. 9.

The display has four main menus:

Status	Indication of system status
Operation	Change of operating parameters such as setpoint
Alarm	Alarm log for fault finding
Settings	Change of settings (password option)

7.1.2 Top line

The top line (B) is illustrated in fig. 9. It shows the following:

- the display number and title (left side)
- the selected menu (left side)
- the symbol $\otimes$ in case of alarm (right side)
- the symbol $\triangle$ in case of warning (right side)
- the symbol $\nearrow$ if the service language has been selected (right side).

7.1.3 Graphical illustration

The graphical illustration (D) may show a status, an indication or other elements, depending on the position in the menu structure. The illustration may show the entire system or part of it as well as various settings.

7.1.4 Scroll bar

If the list of illustration elements exceeds the display, the symbols $\blacktriangle$ and $\blacktriangledown$ appear in the scroll bar to the right. Move up and down in lists with these symbols.

7.1.5 Bottom line

The bottom line (C) shows the date and time.